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Fracture network geometry and injection-induced seismic hazard in engineered subsurface reservoirs

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Geometrical features of fractures such as size (length) distribution exert fundamental control on the seismic-thermal-hydraulic-mechanical processes in the Earth's crust. However, characterization of fracture network at depth remains a major challenge. This limits placing constraints on the statistical relationship between fracture network and injection-induced seismic hazard during underground fluid injection operations. Here, we propose a methodology to integrate borehole data, outcrop analogues and computer simulations to constrain the fractal Discrete Fracture Network (DFN) model of the target rock mass. The method allows a direct comparison with recorded microseismic events. We applied the proposed methodology to the Basel Enhanced Geothermal System (EGS), Switzerland, and found that rupture size distribution exhibits higher power-law exponents than fracture size distribution. This can indicate that larger fractures are prone to partial rather than complete rupture due to pore-pressure increase. To quantify hazard, we developed a novel maximum possible magnitude M_{max} model, combining a dual power-law DFN model with the characteristic length of pore-pressure diffusion. The approach assumes a conservative, worst-case scenario in which the largest critically stressed fault governs the seismic hazard. The derived formulation can be calibrated to any specific site, provided that the fracture network and the hydrogeological conditions of the subsurface system are well-characterized. To account for parameter uncertainties, Monte Carlo simulations can be performed to estimate both the possible range and the most probable value of M_{max}. The methodology was successfully tested on the Basel site, providing M_{max} of 3.4, in a close agreement with the observed M_w 3.2. This framework provides a physics-based tool for seismic hazard assessment in engineered subsurface projects including EGS, geological CO_2 and Hydrogen storage.

KEYWORDS
discrete fracture networks, fracture network characterization, geothermal reservoirs, induced seismicity, seismic hazard

1 Introduction

Fractures and faults are structural discontinuities that influence the thermal–hydraulic–mechanical behavior of Earth's crust. They can act as conduits or barriers for fluid flow, control the mechanical deformation and affect the heat transfer across geological formations (Viswanathan et al., 2022; Sun et al., 2020; Gottron and Henk, 2021; Mahmoodpour et al., 2022). Moreover, they form complex geometries that play a fundamental role in the nucleation and propagation of both natural and induced seismicity (Lee et al., 2024; Scholz, 2019). Induced earthquakes are critical for emerging geoenergy technologies such as Enhanced Geothermal Systems (EGS), Carbon Capture and Storage (CCS) and Hydrogen Storage (Cheng et al., 2023; Ge and Saar, 2022). These earthquakes pose potential risk to infrastructure and reduce public acceptance of these technologies. Hence, identifying the geological and operational controls on induced seismic hazard is essential for engineering subsurface reservoirs.

High-pressure fluid injection is typically performed to enhance the permeability of less conductive formations and enable the extraction of geothermal resources (Jia et al., 2022), as well as to store CO_2 or Hydrogen in the subsurface. However, fluid injection can reactivate pre-existing fracture system and generate seismic slip (Moein et al., 2023). The magnitudes of induced earthquakes are closely linked to the rupture dimensions of the activated fracture planes (Wells and Coppersmith, 1994). Because the rupture dimensions of induced earthquakes are bounded by the size of pre-existing fractures, a detailed understanding of fracture network geometry is critical for seismic hazard assessment.

Despite the importance, fracture network characterization at depth remains a challenging task (Medici et al., 2023). Surface-based geophysical methods suffer from limited resolution, particularly in crystalline basement rocks, where seismic reflectors are poorly defined (Eaton et al., 2003; Villamizar et al., 2022). The only direct measurements of deep fracture system comes from borehole image logs (Wood, 2024). These logs determine the location, orientation and aperture of fractures along the boreholes. However, boreholes sample only a small fraction of the fractures actually occurring in a particular volume. Consequently, the geometrical attributes of 3D fracture network such as density, orientation, length and spatial distribution must be statistically described.

Discrete Fracture Network (DFN) models offer a framework to represent 3D fracture geometry constrained by 1D and 2D datasets and stereological relationships (Moein et al., 2019; Lepillier et al., 2020; Xiao et al., 2025). Since 1D fracture datasets derived from boreholes cannot constrain the fracture length distribution, outcrop analogue studies are executed to provide complementary information on the length distribution (Bisdom et al., 2014; Bauer et al., 2017; Allgaier et al., 2023; Bruna et al., 2019; Laux and Henk, 2015).

Fracture system in geological formations is often self-similar across multiple scales and their spatial and length distributions frequently follow power-law scaling (Bonnet et al., 2001; Torabi and Berg, 2011). In particular, the frequency size distribution of fractures is given by Equation 1:

$$N_{\geq l} = c\, l^{-a_{2D,3D}} \tag{1}$$

where, c is a constant and $a_{2D,3D}$ are the fracture length exponent in 2D or 3D. The symbols used throughout the manuscript are summarized in Table 1. By adopting a power-law perspective, stereological relationships can constrain the three-dimensional geometry of fractures, providing a first-order model for connectivity, density and spatial arrangement within the 3D rock volume (Darcel et al., 2003; Wang, 2005). A power-law model offers the following advantages: First, it has shown a remarkably good fit to the observed fracture data in a wide range of scales (Bonnet et al., 2001; Torabi and Berg, 2011). Second, it is scale-invariant without any characteristic length scale. This enables direct upscaling of fracture network at larger scale from smaller scale observations (Bonnet et al., 2001; Torabi and Berg, 2011). Third, it is consistent with the concept of self-organized criticality, in which scale-invariance emerges in systems at or near critical state (Renshaw and Park, 1997; Sornette and Davy, 1991). Finally, it is widely used for earthquake statistics, reflecting the scale-invariant nature of energy release along faults.

The magnitude distribution of earthquakes is often defined by Gutenberg–Richter (GR) law (Equation 2):

$$N_{\geq M} = 10^{a-bM} \tag{2}$$

where, $N_{\geq M}$ counts the number events exceeding the magnitude M, with a and b being GR constants (Gutenberg and Richter, 1942).

Although the correspondence between fault network and natural earthquakes has been previously studied (Scholz, 2025; Zou and Fialko, 2024; Truttmann et al., 2025), this relation remains poorly investigated for induced earthquakes. Moreover, the existing approaches to constrain the $M_{\max}$ often rely on simplified assumptions and site-specific parameters, controlled by fluid injection volume (McGarr, 2014; Bommer and Verdon, 2024; Shapiro et al., 2010). Nevertheless, these models do not explicitly consider the influence of the underlying fracture system on the eventual rupture size of the largest event. This study fills these gaps by integrating borehole data, outcrop analogues and computer simulations.

Here, we constrain the statistical attributes of subsurface fracture network and compare with observed induced seismicity from Basel EGS in the Upper Rhine Graben (northern Switzerland). Building upon the DFN reconstruction, we introduce a novel $M_{\max}$ model that explicitly integrates three-dimensional fracture network geometry, providing a geological assessment of seismic hazard. Eventually, we apply the proposed model for the Basel site and evaluate the impact of parameter uncertainties on the $M_{\max}$ estimate in real field conditions.

2 Methods

2.1 Constraining a statistical DFN model

We apply a statistical DFN model, assuming a power-law distribution of fracture length and fractal organization of fracture centers (fractal spatial distribution) (Davy et al., 1990), defined by Equation 3:

$$n(l,L)\, dl = \gamma\, L^{D_{3D}} l^{-a_{3D}}\, dl \tag{3}$$

where, $n(l,L)\,dl$ counts the fractures whose centers are in a cuboid of size L with lengths between l and $l+dl$. The parameter γ is a normalization constant affecting the fracture count, D_{3D} is the correlation dimension of fracture centers and a_{3D} is the fracture size exponent. Based on the methodology developed by Moein et al. (2019), setting up a statistical DFN model requires the following input parameters: D_{3D}, a_{3D}, γ and minimum fracture length $l_{\min}$ (used as a cut-off for the length distribution). Despite the inherent uncertainties, D_{3D} and a_{3D} can

TABLE 1 List of symbols.

Symbols	Description
a_{2D} *and* a_{3D}	Fracture length (size) exponent for 2D outcrops and 3D space
$N_{\geq l}$	Number of fractures exceeding length l
a	a-value of Gutenberg-Richter law
b	b-value of Gutenberg-Richter law
$N_{\geq M}$	Number of earthquakes exceeding magnitude M
$\gamma\ [m^{a_{3D}-D_{3D}-1}]$	Normalization constant of dual power-law model
$L\,[m]$	Domain length in 1D, 2D and 3D
D_{1D}	Correlation dimension of fracture intersections along 1D boreholes
D_{2D} *and* D_{3D}	Correlation dimension of fracture centers in 2D outcrops and 3D space
$N_p(\frac{r}{L})$	Number of pairs of fractures whose center-to-center normalized distance is less than $\frac{r}{L}$
$C_2(\frac{r}{L})$	Normalized correlation integral NCI
$r\,[m]$	Distance between fracture centers or intersections
N_t	Total number of fractures
$l_{min}\,[m]$	Minimum fracture length in DFN generation
$w\,[m]$	Fracture width (opening)
$D\,[m^2]$	Hydraulic diffusivity
$T\,[s]$	Elapsed time from the start of injection
M_w^{max} or M_{max}	Maximum possible magnitude
M_w	Moment magnitude
M_0	Seismic moment
R_{max}	Maximum rupture size
$\Delta\sigma\,[MPa]$	Stress drop
Ψ	Fracture network seismogenic potential
$T_{inj}\,[s]$	Duration of injection protocol
β	Rupture-size exponent

be estimated from lower-dimensional datasets using stereological relationships (Darcel et al., 2003).

The spatial distribution can be constrained by 1D borehole fracture datasets, using a Normalized Correlation Integral NCI function of fracture intersections (Moein et al., 2022; Wang et al., 2019; Marrett et al., 2018). A power-law can fit the NCI according to Equation 4:

$$C\left(\frac{r}{L}\right) = \frac{2}{N_t(N_t-1)} N_p\left(\frac{r}{L}\right) \sim \left(\frac{r}{L}\right)^{D_{1D}} \quad (4)$$

with N_t as the total number of fractures and $N_p(\frac{r}{L})$ counting the pairs of fracture centers whose normalized distance is less than $\frac{r}{L}$. The correlation dimension D_{1D} is obtained from the slope of $C\,(\frac{r}{L})$ versus $\frac{r}{L}$ in log–log space. For fractal datasets, NCI displays linear behavior in log-log space. Deviations from linearity occur at large scales due to boundary effects and at small scales (0.001 normalized units, corresponding to 1–3 m) due to undersampling of closely spaced fractures. The correlation dimension is typically determined by fitting a linear trend to the NCI over the normalized range 0.001–0.1, with uncertainty quantified as the standard deviation of the local slope estimates within this range. Analyses of 1D fracture datasets indicate that the correlation dimension is typically ranges between 0.7 and 0.9 (Moein et al., 2022; Brixel et al., 2020; Du Bernard et al., 2002). The first-order approximate of D_{3D} in fractal DFNs is computed by $D_{3D} = D_{1D} + 2$ (Darcel et al., 2003). These relations are valid for statistically isotropic and self-similar fracture networks. Moreover, the uncertainty in D_{1D} will be similar to the D_{3D}, following the standard uncertainty propagation.

Since 1D fracture datasets cannot provide any information on the length of fractures, outcrop datasets are used to constrain the 3D fracture length distribution. A first-order approximation of a_{3D} is typically computed by the following scaling relationship $a_{3D} = a_{2D} + 1$ (Darcel et al., 2003). The reported values of fracture length exponent in outcrops is within the range of 1.5–3.5 (Lei et al., 2017). Note that, the lower exponents indicate networks dominated by larger fracture, whereas higher exponents indicate the dominance of shorter ones.

The parameter γ cannot be directly constrained by 1D or 2D fracture datasets. To overcome this limitation, we propose estimating γ from synthetic fractal DFN models, calibrated to the linear fracture intensity P_{10} (fracture count per unit length) derived from borehole image logs. Sufficient number of synthetic DFNs (e.g., 1,000 realizations in a 1,000 m cubic domain) with varying a_{3D} can be generated. Since the DFNs follow a fractal model, the resulting relationships are scale-independent and applicable to smaller or larger domains. Then, virtual wells can be placed at the center of each DFN realization to compute the resulting P_{10} values. For a given D_{3D}, it is possible to establish a relationship between P_{10}, γ and a_{3D}.

The value of l_{min} can be constrained by the resolution limits of borehole image logs, which only detect fractures above a certain aperture. A rough estimate of an appropriate minimum fracture length can be obtained from aperture–length scaling relationships in fracture mechanics (Olson, 2003) or outcrop analogue studies (Ortega et al., 2006). If the minimum fracture length is significantly smaller than the largest fracture ($l_{min} \ll l_{max}$), the total number of fractures exceeding the minimum length $N_{\geq l_{min}}$ can be computed using Equation 5.

$$N_{\geq l_{min}} = \int_{l_{min}}^{\infty} n\ (l, L)\ dl = \frac{\gamma L^{D_{3D}}}{a_{3D}-1}\ l_{min}^{1-a_{3D}} \quad (5)$$

2.2 Geometrical M_{max} model

Building upon the statistical properties of the fracture network, we develop a new M_{max} model for injection-induced seismicity. It is reasonable to assume that the largest induced earthquake likely occurs on the largest critically-stressed fault in a perturbed rock volume. Then, we derive the largest expected fracture length within a cuboid size L, based on the fractal statistics. The schematic illustration of cumulative 3D fracture size distribution is shown in Figure 1 and the largest possible length can be derived using the Equation 6. Note that the slope of complementary cumulative length distribution in 3D space is $a_{3D} - 1$ (Bonnet et al., 2001).

$$\log\ (l_{max}) = \log\ (l_{min}) + \frac{1}{a_{3D}-1} \log\ \left(N_{\geq l_{min}}\right) \quad (6)$$

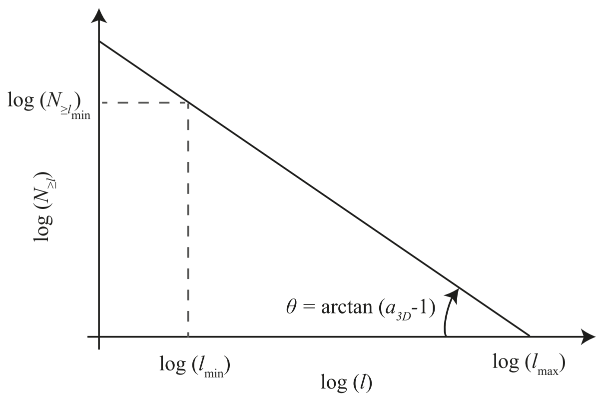

FIGURE 1
Complementary cumulative 3D fracture size distribution following a dual power-law model with an exponent of a_{3D}. l_{min} indicates the minimum and l_{max} represents the maximum fracture length. The total number of fractures exceeding the minimum length $N_{\geq l_{min}}$ can be derived from Equation 5.

Constraining the size of largest fracture length l_{max} requires the definition of domain size. This size can be approximated by the perturbed rock volume by pore-pressure diffusion caused by fluid injection. This is often estimated by the concept of triggering front as in Equation 7.

$$L = \sqrt{4\pi DT} \tag{7}$$

Here, L is characteristic length-scale of pore-pressure diffusion, D $[m^2/s]$ is the hydraulic diffusivity and T $[s]$ is the elapsed time from the start of injection (Shapiro, 2015). This represents a heuristic radius of disturbance and estimates the temporal propagation of pore-pressure front. Combining Equations 5–7 gives an estimate of l_{max} as in Equation 8.

$$\log\ (l_{max}) = \log\ (l_{min}) + \frac{1}{a_{3D}-1}\left(\log(\gamma) + \frac{D_{3D}}{2}\log(4\pi D) + \frac{D_{3D}}{2}\log(T) + (1-a_{3D})\log(l_{min}) - \log(a_{3D}-1)\right) \tag{8}$$

The maximum seismic moment M_0^{max} can be computed using the maximum fracture length $R_{max} = \frac{l_{max}}{2}$ following the well-known seismological relation (Equation 9) (Eshelby, 1957):

$$M_0^{max} = \frac{16}{7}\ (R_{max})^3\ \Delta\sigma \tag{9}$$

where, $\Delta\sigma$ is the stress drop. The resulting M_0^{max} can be expressed as the Equation 10:

$$M_0^{max} = \frac{2\,\Delta\sigma\, l_{min}^3}{7} \times 10^{\frac{3\left(\log(\gamma)+\frac{D_{3D}}{2}\log(4\pi D)+\frac{D_{3D}}{2}\log(T)+(1-a_{3D})\log(l_{min})-\log(a_{3D}-1)\right)}{a_{3D}-1}} \tag{10}$$

The moment magnitude is also related to the seismic moment M_w using Equation 11 (Kanamori and Anderson, 1975):

$$M_w = \frac{2}{3}\ (\log_{10}(M_0) - 9.1) \tag{11}$$

If the M_0^{max} is integrated into the previous equation, the resulting M_w^{max} can be written as Equation 12.

$$M_w^{max} = \frac{D_{3D}}{a_{3D}-1}\log(T) + \frac{2}{3}\log\ \left(\frac{2\Delta\sigma}{7}\right) + 2\log\ (l_{min}) + \frac{2}{a_{3D}-1}\left(\log\ (\gamma) + \frac{D_{3D}}{2}\log\ (4\pi D) + (1-a_{3D})\ \log\ (l_{min}) - \log\ (a_{3D}-1)\right) - 6.06 \tag{12}$$

Here, we define a new parameter, referred to as fracture network seismogenic potential Ψ, which quantifies the inherent capacity of a fractured rock to generate seismic rupture using Equation 13.

$$\Psi = \frac{2}{3}\log\ \left(\frac{2\Delta\sigma}{7}\right) + \frac{2}{a_{3D}-1}\left(\log\ (\gamma) + \frac{D_{3D}}{2}\log\ (4\pi D) - \log\ (a_{3D}-1)\right) - 6.06 \tag{13}$$

The parameter Ψ quantifies the joint impact of fracture network properties (including length distribution, spatial organization and density), hydraulic diffusivity and stress drop on M_{max}. Accordingly, the ultimate M_{max} model can be expressed by Equation 14.

$$M_{max} = \frac{D_{3D}}{a_{3D}-1}\log\ (T) + \Psi \tag{14}$$

2.3 Probabilistic M_{max} estimation

It should be noted that the proposed model (Equation 14) provides only a geometric upper bound of M_{max} in a fractured rock if an event occurs, without providing the probability of occurrence. Given the inherent uncertainties of the DFN parameters, stress drop and hydraulic diffusivity in the Earth's crust, the expected M_{max} must be estimated using a probabilistic approach. This can be achieved by a Monte Carlo simulation that can estimate the most likely M_{max}. By explicitly incorporating parameter uncertainties, this approach provides a more realistic evaluation of seismic hazard.

The DFN parameters (γ, D_{3D} and a_{3D}) can randomly sampled from a uniform distribution within a physically plausible range for target rock mass. These parameters are constrained using borehole observations and outcrop analogues, which commonly exhibit scale-invariant behavior over multiple orders of magnitude.

Additional uncertainties arise through variability of stress drop ($\Delta\sigma$) and hydraulic diffusivity (D). Stress drop is typically modeled using a normal distribution centered at 1 MPa with a standard deviation of 0.1 MPa, consistent with typical estimates for induced earthquakes (Goertz-Allmann et al., 2011). Moreover, hydraulic diffusivity can also sampled from a uniform distribution within a broad range of crustal conditions. All parameters are randomly sampled and combined in a large ensemble of realizations, with each realization producing an independent estimate of M_{max}. Repeating this procedure for a large number of realizations yields a probability distribution of M_{max}, from which the most probable value and associated uncertainty can be quantified.

3 Application

3.1 Geological setting and fracture dataset

In this section, we demonstrate the application of the new M_{max} model to a real EGS reservoir. Here, we select the Basel EGS, located in the southern part of Upper Rhine Graben area, which is recognized for its geothermal resources in central Europe. In 2006, BS–1 well was drilled to a depth of 5 km beneath the city of Basel, Switzerland. The borehole was imaged by an Ultrasonic Borehole Imager (UBI) with a vertical resolution of approximately 1 cm (Valley and Evans, 2009). A detailed interpretation of the UBI image logs is presented in previous studies (Ziegler et al., 2015; Ziegler and Evans, 2020). Here, we use the same dataset and a summary of the methodology is presented. Fracture traces were picked by fitting sinusoids to the unwrapped UBI images. The primary filtering and removal rule distinguished natural fractures from drilling-induced features. Full sinusoidal traces were classified as certain natural fractures and incomplete traces were marked as uncertain. Approximately 75% of the selected features were classified as certain and the rest were marked uncertain. Here, we retained both categories to maximize data coverage and improve the statistical robustness. These features were identified by specific patterns such as axial or en-echelon alignments with regular spacing (0.1–0.3 m). Poor-quality images due to tool stick-slip motion below 4,677 m were corrected using accelerometer-derived displacement estimates. Missing or spurious data in the borehole cross-sectional area log were interpolated between adjacent points. Fracture orientations (dip and strike) derived from UBI images were corrected for borehole deviation, which reached up to 8° from vertical.

The crystalline basement was encountered at a depth of approximately 2.6 km (Figure 2A), consisted of plutonic rocks with no metamorphism. In Figure 2A, the fracture counts were identified using a non-overlapping depth interval of 1 m, with edges starting from 2.6 km. The binning was arbitrary and chosen for visualization purposes. The uppermost 480 m is densely fractured, with up to nine fractures per meter. This highly fractured interval is interpreted as a paleoweathered surface, which was exposed to surface processes before burial. This interval also shows significant alteration, including mineralogical and structural changes. For computing the NCI, the entire dataset was used. Since there is no depth dependence of fractal dimension, this choice does not affect the results and increases the robustness of correlation analyses (Moein et al., 2022). For the computation of P_{10} (for constraining the 3D DFN in the next section), the first 480 m is excluded from the analyses. Figure 2B displays the orientations of 1,164 fractures detected between 2.6–5 km, plotted as poles on a lower-hemisphere, equal-area stereonet.

3.2 Constraining the 3D DFN geometry

Despite the fracture spatial and length distributions (which can be estimated from 1D and 2D fracture datasets), there is not a standard methodology to constrain the parameter γ in 3D space. To overcome this limitation, we follow the proposed methodology in Section 2.1. The hypothesis is to estimate γ synthetic fractal DFNs calibrated to the linear fracture intensity P_{10} (fracture count per unit length) derived from borehole image logs.

We generate synthetic DFNs within a 1,000 m cubic domain to establish a relationship between P_{10}, γ and a_{3D}. The detailed methodology for generating synthetic DFNs has been described in Moein et al. (2019). The domain size is set to 1,000 m, that is comparable to the size of an EGS. No geometrical constraint was implemented on the boundaries (i.e., the fractures could propagate outside of the study domain). Fracture orientation was randomly selected and the minimum fracture length was set to 10 m. This selection is arbitrary and reflects the resolution limits of borehole image logs, which only detect fractures above a certain aperture. However, the same procedure can be repeated for any other cutoff value. A rough estimate of an appropriate minimum fracture length can be obtained from aperture–length scaling relationships in fracture mechanics. Fractal spatial distribution is generated Multiplicative Cascade process, which creates both scale-invariant fracture patterns. Since the DFNs follow a fractal model, the resulting relationships are scale-independent and applicable to smaller or larger domains. Virtual wells are placed at the center of each DFN realization to compute the resulting P_{10} values.

Fracture length exponents for various rock types is generally expected to fall within the range of $2.5 \leq a_{3D} \leq 4.5$. However, this exponent is often between three and four in crystalline basement rocks (Lavoine et al., 2024). Also, the fractal dimensions of 1D fracture datasets in deep boreholes is between 0.8–0.9 (Moein et al., 2022). Figure 2C shows the correlation analysis of fractures in the BS–1 well. We computed the NCI of fracture using 100 logarithmically spaced normalized distance bins between 0.0001 and 1.0. This results in 25 data points per decade. Local slopes were determined by performing linear fits over consecutive windows of five points without overlap, producing five pairs of slope values per decade. The final values of D_{1D} were obtained as the mean of the slopes within the normalized distance range. Error estimates for D_{1D} were computed as the standard deviation of the 15 local slopes within this range. Note that the full dataset was used (between 2.6–5 km) without any further processing. Geiven the $D_{3D} = 2.85$, we generate 100 DFN realizations for five representative exponents of 2.5, 3.0, 3.5, 4.0 and 4.5.

Figure 3 illustrates the resulting dependence of γ on P_{10} for the different fracture length exponents. The mean P_{10} was computed and the error bars represented the range of P_{10} for 100 realizations. Linear regression was applied to determine the slope P_{10} versus γ. Subsequently, the dependency of the slope m on a_{3D} was fitted in log-space using a linear function. These results provide a practical means for constraining fracture abundance within the 3D reservoir volume based on borehole observations. Based on the trends shown in Figure 3, the parameter γ depends on fracture length exponent, fracture intensity and the correlation dimension of fracture centers. Equation 15 can be applied to compute the parameter γ for a given correlation dimension of $D_{3D} = 2.85$.

$$\gamma = \frac{P_{10}}{10^{-1.46\,a_{3D}+4.51}} \qquad \text{for} \quad D_{3D} = 2.85 \tag{15}$$

3.3 Fractures and rupture size distribution

In 2006, hydraulic stimulation was performed in the open-hole section of the BS-1 well (between 4,629 and 5,000 m). The injection operations lasted approximately 6 days and the associated

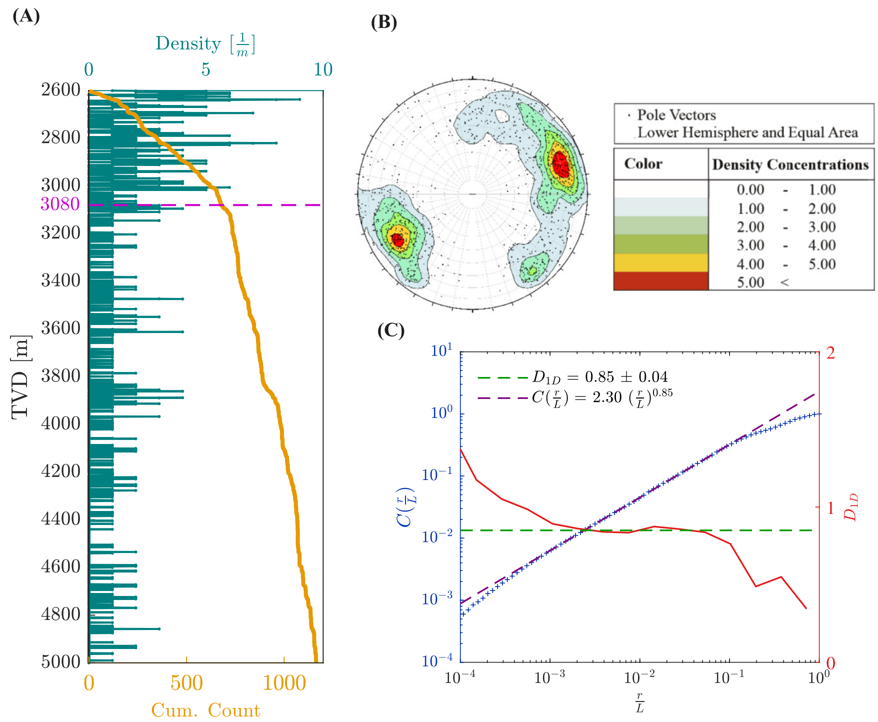

FIGURE 2
(A) Fracture density (number of fractures per 1 m intervals) and cumulative fracture count versus depth in the BS-1 well (Crystalline basement). The first 480 m (up to a depth of 3,080 m) has a significantly higher fracture density. **(B)** Equal-area stereonet projection of 1,164 fracture poles on a lower-hemisphere, equal-area stereonet. **(C)** Two-point correlation function $C(\frac{r}{L})$ (blue stars) and its local slope (red curve) of BS-1 fracture dataset. The purple dashed line represents the power-law fit and the dashed green line shows the estimate of D_{1D} within the normalized distance of 0.001 and 0.1.

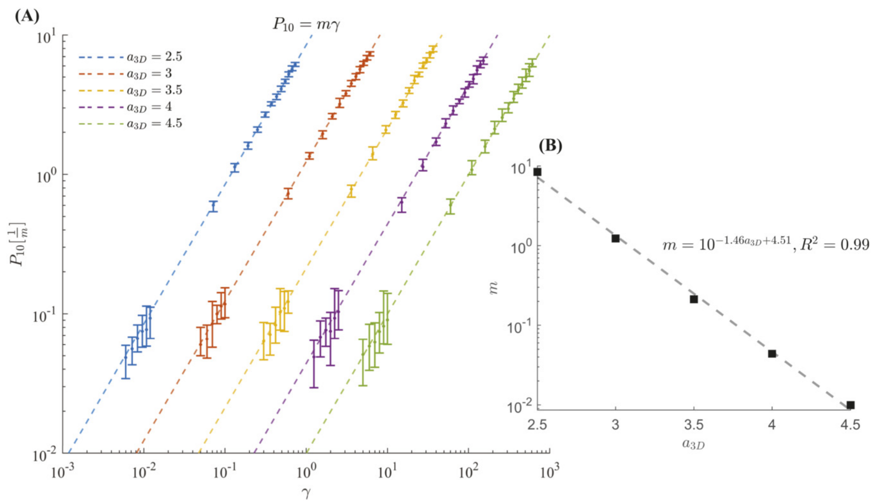

FIGURE 3
Constraining the parameter γ using P_{10} for different fracture length exponents a_{3D} = 2.5, 3.0, 3.5, 4.0 and 4.5 and D_{3D} = 2.85. **(A)** Error bars indicate the range of possible P_{10} for 1000 DFN realizations as a function of γ. The dashed lines are linear fit to the average P_{10} and the slope m is a function of fracture length exponent. **(B)** The relation between calculated m and fracture length exponent a_{3D}.

microseismicity was analyzed using a waveform similarity relocation method (Kraft and Deichmann, 2014). Also, it is known that during the shut-in phase, the pressurized zone continues to expand for a period similar to the injection duration (Johann et al., 2018). Since the injection halted after 6 days, the pore-pressure front is expected to continue increasing for a total of approximately 12 days. Within this period, nearly 1,800 events were recorded with magnitudes exceeding the magnitude of completeness $M_c = 0.8$

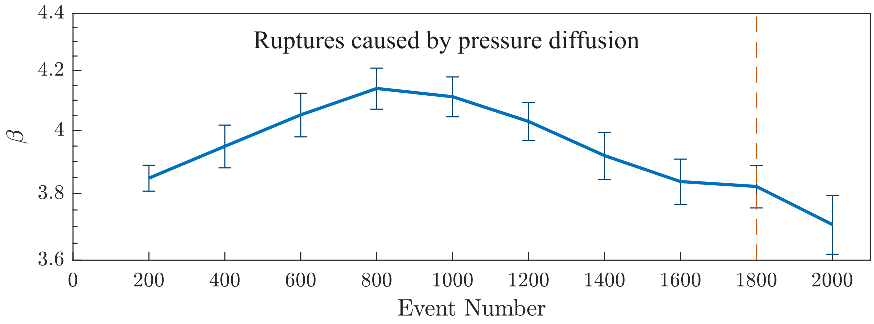

FIGURE 4
The temporal evolution of rupture size exponent β in the Basel EGS. Each window contains 200 events exceeding the magnitude of completeness M_c = 0.8. The first 1800 events were reported in the first 12 days and are likely dominated by pore-pressure diffusion. The rupture radii are computed using a widely-accepted stress drop of $\Delta\sigma = 1$ MPa.

(Moein et al., 2018). Figure 4 shows the temporal evolution of rupture size exponent β during the first 12 days and onwards. The rupture radius was computed using a widely accepted stress drop of $\Delta\sigma = 1$ MPa (using Equations 10 and 12) and the exponent was derived by the maximum likelihood fit (Aki, 1965). The magnitude of completeness was set to 0.8 using the Maximum Curvature Method (Moein et al., 2018). Robustness checks were performed by varying the magnitude of completeness by ±0.1 and the resulting rupture-size exponents remained within the estimated uncertainty bounds, indicating that the results are stable. The derived exponents are between 3.75 and 4.25, which are higher than the fracture length exponent derived from lower-dimensional data such as outcrops $a_{3D} = 3$.

Although there is no direct outcrop analogue study at the Basel site, several independent investigations in the broader Upper Rhine Graben region provide relevant insights. Multiscale analyses of fracture system in this region have shown that the fracture length distribution typically follows an exponent of approximately 2 (Bossennec et al., 2021; Frey et al., 2022). Although the analogues studies were approximately 300 km away from the Basel site, both sites belonged to the same Variscan basement unit. These units typically were formed under similar tectonic history and the fracture systems are comparable. Under the assumption of fractal model, the 3D power-law size distribution is expected to have an exponent of $a_{3D} = 3$.

3.4 Geometrical M_{max} estimate and uncertainty analysis

Starting with spatial distribution, $D_{1D} = 0.85 \pm 0.04$ provided a first-order estimate of 3D fractal dimension as $D_{3D} = 2.85 \pm 0.04$. Hence, the normalization parameter γ can be constrained using the simulation-based Equation 15 for the possible range of a_{3D}. Here, we set $l_{\min} = 10$ m and consider the possible range of $3 \leq a_{3D} \leq 4$ for the crystalline basement rocks. The stress drop values of induced earthquakes are expected to vary between 0.1 and 10 MPa. However, most of the stress drop estimates are clustered around an average value in the order of 1 MPa (Goertz-Allmann et al., 2011). In this study, we adopt a commonly used normal distribution with an average $\Delta\sigma = 1$ MPa and standard deviation of 0.1 MPa. Additionally, the hydraulic diffusivity in the Earth's crust often ranges between 0.001 and 0.1 ($\frac{m^2}{s}$) (Moein et al., 2026). All parameters, except stress drop which follows a normal distribution, were sampled from a uniform distribution.

Given these ranges, we performed a Monte Carlo simulation consisting of 10^6 realizations. All parameters required $M_{\max}$ estimates were randomly sampled from the feasible ranges. Convergence diagnostics were also performed by assessing the stability of $M_{\max}$ estimates versus the number of realizations. The simulations indicate that the most likely value of the fracture network seismogenic potential is −3. Fitting a normal distribution to the resulting $M_{\max}$, the most probable magnitude is 3.4 ± 0.5 (Figure 5). This magnitude is a close agreement with the observed $M_w = 3.2$ at the Basel site, which was recorded several hours after shut-in.

4 Discussion

4.1 Comparison with other M_{max} models

Maximum possible magnitude is a key metric for seismic hazard and risk assessment of industrial activities. Classical volume-based models often consider injection volume as the main source of strain (McGarr, 2014; Yu et al., 2024; Galis et al., 2017). While these approaches capture some aspects of fluid-induced seismicity, they are limited in seismically active regions where tectonic stress significantly contributes to fault rupture (Equation 16).

Shapiro et al. (2021) introduced a time-based $M_{\max}$ scaling, linking magnitude directly to earthquake triggering time as:

$$M_{\max} \sim 2\,\xi \log(T) \tag{16}$$

where, ξ accounts for the physical process and T is the triggering time. They observed a direct relation between $M_{\max}$ and the elapsed time from the start of injection for many case histories. They showed that the earthquake triggering time is more indicative of the underlying physical mechanism and can better replace the injection volume. The parameter ξ represents of the physical mechanism, being $\frac{1}{2}$ for pore-pressure diffusion and $\frac{1}{3}$

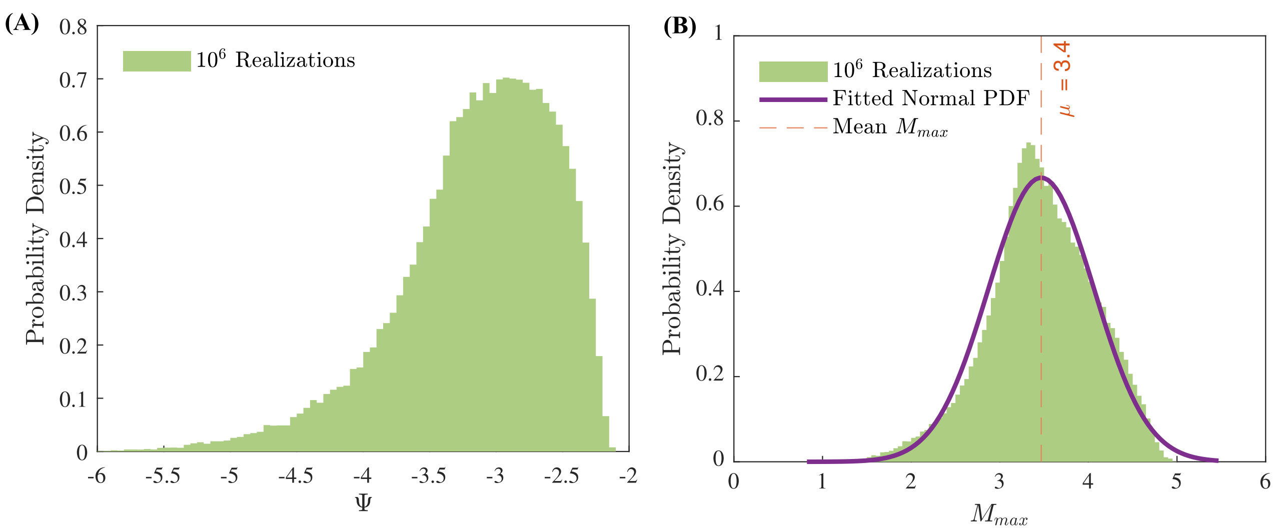

FIGURE 5
Application of geometrical M_{max} model to the Basel EGS. One million realizations were generated by sampling from plausible values of fracture normalization parameter γ, spatial distribution D_{3D}, size distribution a_{3D}, stress drop $\Delta\sigma$ and hydraulic diffusivity D. **(A)** Distribution of fracture network seismogenic potential with the most likely value of −3. **(B)** Distribution of M_{max} fitted with normal distribution. The average M_{max} value is 3.4 with a standard deviation of 0.5.

$\leq \xi < \frac{1}{2}$ for a non-linear diffusion process. A simple comparison with the proposed model (Equation 14) shows that the parameter ξ is comparable to $\frac{D_{3D}}{2(a_{3D}-1)}$ in a fracture system. Using the value of $\frac{D_{3D}}{2(a_{3D}-1)}$, one can also determine the governing physical processes in a DFN.

Further development of the time-based M_{max} model considers the case when the rupture size is on the order of the characteristic length of pore-pressure diffusion (Langenbruch et al., 2024). In this scenario, the nucleation model predicts the M_{max} using Equation 17:

$$M_{\max} = \log(T) + \theta_{D\sigma} \tag{17}$$

where $\theta_{D\sigma}$ is the seismic nucleation constant. This site-specific parameter represents the characteristic time over which subsurface conditions permit the occurrence of an earthquake, effectively quantifying the maximum magnitude for a triggering time of one second. The nucleation constant is defined as $\theta_{D\sigma} = \log(D) + \frac{2}{3}\log(\Delta\sigma) - 4.73$, where D denotes the hydraulic diffusivity and $\Delta\sigma$ refers to the average stress drop. In this relation, $\theta_{D\sigma}$ does not directly include any information on the fracture network geometry. This formulation is a special case of the new M_{max} (Equation 14), where $a_{3D} = D_{3D} + 1$. This implies that the fracture network is self-similar (Bour et al., 2002) and the minimum fracture length does not have an impact on the geometrical M_{max}.

Since the triggering time of an induced earthquake is not known, the injection duration (T_{inj}) can be used as a practical proxy in Equation 17 (Moein et al., 2026). This substitution enables a straightforward comparison of different injection protocols for a given fluid volume, providing a simple seismic hazard assessment at the end of injection (Equation 18).

$$M_{\max} = \log\left(T_{inj}\right) + \theta_{D\sigma} \tag{18}$$

For practical applications, Equation 14 can be written in the form of Equation 19.

$$M_{\max} = \frac{D_{3D}}{a_{3D}-1}\log\left(T_{inj}\right) + \Psi \tag{19}$$

One major advantage of the geometrical M_{max} model is allowing the ruptures to extend beyond the perturbed rock volume. In earlier formulations of the time-based model, rupture could extend outside the pressurized region, only when a sufficiently large area of a pre-existing fault was perturbed (Langenbruch et al., 2024; Shapiro et al., 2021). In the new formulation, ruptures can extend beyond the perturbed region whenever the center of a pre-existing fault is located inside the perturbed domain. Under this geometric criterion, the entire length of a critically-stressed fault can rupture, provided that its center is sufficiently perturbed. As illustrated in Figure 6, fractures A and B can generate induced seismic events, whereas fracture C does not. This represents an upper-bound scenario, not the expected rupture extent, since real ruptures can stop due to different factors such as stress heterogeneities and geometrical asperities. This upper-bound is derived from the geometric assumptions of fractal DFNs in the dual power-law model, where fractures are statistically represented by their centers and lengths.

4.2 Physical interpretation

Physically, Ψ reflects how these fracture network characteristics control M_{max} in a rock volume perturbed by pore-pressure diffusion. Higher Ψ values correspond to networks with a greater potential for nucleation of larger seismic events.

We applied the proposed framework and geometrical M_{max} theory to the Basel EGS site. We performed a Monte Carlo simulation to account for the parameter uncertainty. By sampling

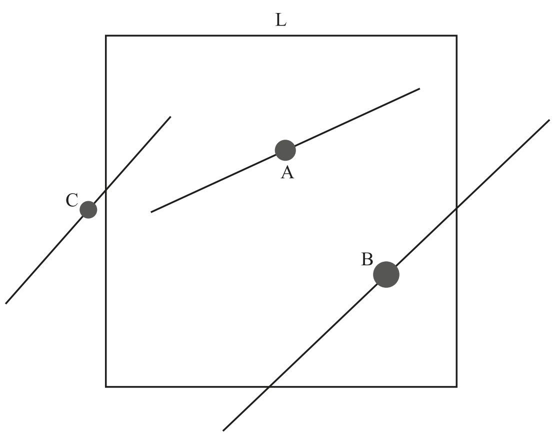

FIGURE 6
A schematic 2D illustration of fractures generating seismicity within a perturbed rock volume of size *L*. Fractures A and B have their centers located inside the perturbed domain and therefore can generate seismic events. Fracture C, whose center lies outside the perturbed region, is not considered in this model. Note that fracture B can extend outside of the perturbed volume.

one million realization, we found the most probable fracture network seismogenic potential of $\Psi = -3$ and a corresponding maximum magnitude of $M_{max} = 3.4 \pm 0.5$, in a good agreement with the observed maximum magnitude of $M_w = 3.2$. Note that the inherent uncertainty of $M_{\max}$ is originated from the properties of 3D fracture network geometry and hydromechanical properties of fractured rocks.

We further analyzed the rupture size distribution in the Basel and found higher exponents than the fracture size exponent. The rupture exponents were in the range 3.7–4.25, higher than the $a_{3D} = 3$, which is inferred from lower-dimensional datasets (outcrop analogues). This discrepancy suggests that larger fractures do not rupture completely during the hydraulic stimulation and partial ruptures deviate the rupture distribution from fracture size distributions. The concept of partial rupture is further elaborated in the discussion with reference to the potential mechanisms. These mechanisms include fault morphology, geometric discontinuities, stress barriers, fault segmentation, material heterogeneity and mechanical anisotropy (Qiu et al., 2016; Manighetti et al., 2007; Wesnousky, 2006; Beroza and Mikumo, 1996; Aki, 1979). Our findings are also consistent with the results from natural earthquakes and fracture network in the Swiss Alps (Truttmann et al., 2025). This study also found that earthquake rupture size distribution shows higher exponents at shallower depths and decreases with depth.

Fracture network geometry in rocks exhibit a multiscale nature, ranging from microcracks to macroscopic faults. Microscale studies reveal that repeated microcracking and crack coalescence can enhance fracture connectivity at mesoscopic scales, providing experimental evidence for multiscale fracture interactions (Sun et al., 2024; Li et al., 2025; Xue et al., 2025). These multiscale fracture network can influence stress heterogeneity and create barriers, preventing the complete rupture of larger fractures.

Not only fracture network geometry, but also a wide range of operational parameters can influence the occurrence of induced seismicity. These parameters include, injection well layout, injection depth, injection duration, reservoir thickness, proximity to large-scale faults and the connectivity can influence the induced seismicity (Roche et al., 2025; Hincks et al., 2018; Boitz et al., 2024; He et al., 2025; Moein et al., 2026).

4.3 Implications and limitations

Note that the application of the proposed framework requires high-quality datasets. High-resolution borehole image logs, dense microseismic monitoring networks and well-characterized outcrop analogues are essential for constraining fracture geometry and validating the inferred scaling relationships. However, such datasets are hardly available in real field cases, which can complicate the practical implementation. This limitation reflects the current availability of suitable data rather than a weakness of the proposed approach.

Moreover, the geometrical $M_{\max}$ applies the concept of triggering front, which considers fluid injection into a virtually infinite volume. This assumption limits the application of the proposed approach to bounded reservoirs, where the layer thickness controls the $M_{\max}$ (Moein et al., 2026). Moreover, the proposed model considers the pore-pressure diffusion as the primary triggering mechanism of induced seismicity. However, various mechanisms such as poroelastic effects, aseismic slip, earthquake interactions and thermoelastic effects (Moein et al., 2023) and incorporating all these mechanisms into the current approach is challenging. Future development of the model can include thermal coupling to evaluate the impact of thermal stresses on the perturbed rock volume and induced seismic hazard. Despite the limitations, the proposed framework is broadly applicable to seismic hazard assessment of engineered subsurface systems, including CCS, Hydrogen Storage and hydrocarbon reservoirs.

5 Conclusion

In this contribution, we present a novel framework that links the 3D fracture network geometry to injection-induced seismic hazard in engineered subsurface reservoirs. By integrating borehole images, outcrop analogue studies and synthetic fractal fracture networks, we constrain the 3D DFN models of the target formations at depth. Moreover, we developed a theoretical $M_{\max}$ model that explicitly incorporates fracture geometry, with uncertainties quantified via Monte Carlo simulations. The proposed model extends the time-based $M_{\max}$ models by allowing ruptures to extend beyond the perturbed volume. Application to the Basel EGS site reproduces observed seismicity, which demonstrate the predictive potential under real field conditions. We also show that rupture size distributions can deviate from fracture statistics due to partial rupture of larger fractures. By linking statistical properties of fracture system with $M_{\max}$, this methodology introduces a physics-based and probabilistic tool for seismic hazard assessment in subsurface applications, including geothermal energy, CCS and Hydrogen Storage projects.

Data availability statement

The microseismic earthquake catalogue for the Basel EGS used in this study is publicly available at https://doi.org/10.6084/m9.figshare.31272376. The borehole image logs are owned by Geo-Energy Suisse and analyzed by Dr. Martin Ziegler at Swisstopo. The data owner retains the full dataset and access can be granted upon reasonable request.

Author contributions

MM: Writing – original draft, Software, Investigation, Writing – review and editing, Formal Analysis, Methodology, Data curation, Conceptualization, Validation, Visualization. KR: Funding acquisition, Writing – original draft, Writing – review and editing, Project administration, Supervision. AH: Project administration, Writing – original draft, Resources, Writing – review and editing, Supervision, Funding acquisition.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. The authors acknowledge the use of ChatGPT for assistance in improving the clarity and readability of the English language in this manuscript.

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